

CanSat Devboard Fabrication Document

Layer Stack Legend

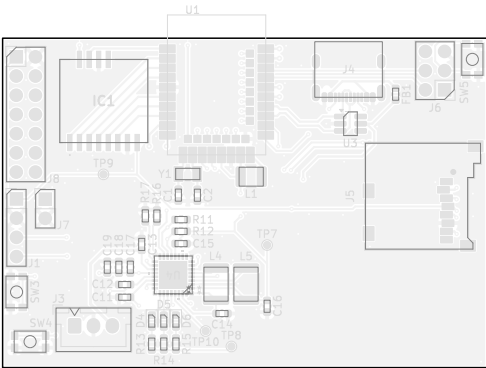
	Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste				Paste Mask	
	F,Silkscreen				Legend	GBR
	F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
	Core		1,21mm	FR4	Dielectric	
	Copper	L3 (GND)	0,035mm (1,00oz)		Plane	GBR
	Prepreg		0,1mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
	B,Silkscreen				Legend	GBR
	B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

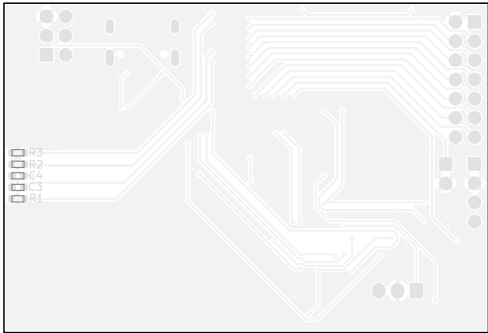
Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



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Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

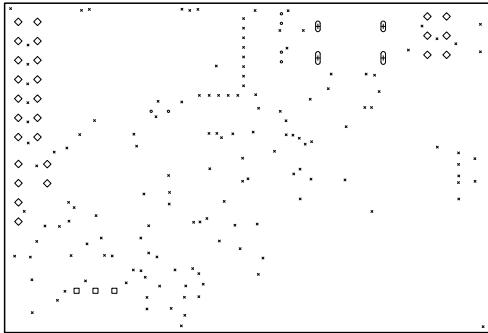
	Comments:	Company: Karpát Space Program		Variant: CHECKED	Git Hash: e9cdcd6
		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	145	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
○	6	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	4	0,60mm (23,62mils)	PTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	3	0,95mm (37,40mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	26	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 184						

Drill Drawing L1 - L4 (Scale 1:1)



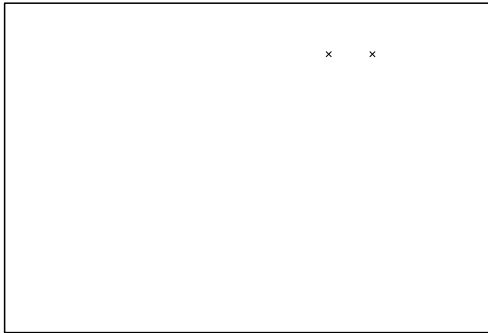
	Comments:	Company: Karpát Space Program		Variant: CHECKED	Git Hash: e9cdcd6
		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	2	0,65mm (25,59mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
	Total 2					

Drill Drawing L1 - L4 (Scale 1:1)



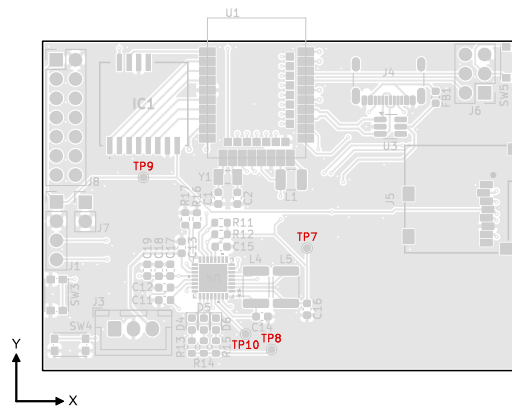
	Comments:	Company: Karpát Space Program		Variant: CHECKED	Git Hash: e9cdcd6
		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP7	VMCU	87.96	1.25
TP8	VSYS	83.23	-12.16
TP9	+3.3V	66.31	10.60
TP10	VLDO	79.83	-10.12

Ref.	Net	X [mm]	Y [mm]
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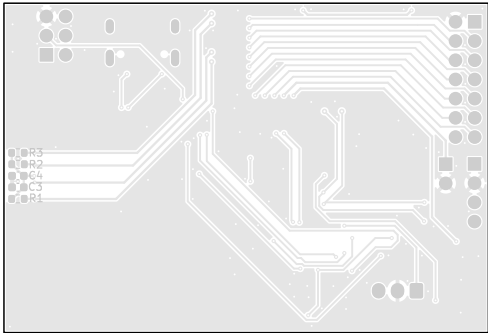
All dimensions are in millimeters unless otherwise specified.

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		Board Name: CanSat Devboard		Project Name: Cansat Devboard			
	Sheet Title: Top Test Points (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)		
	Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10		

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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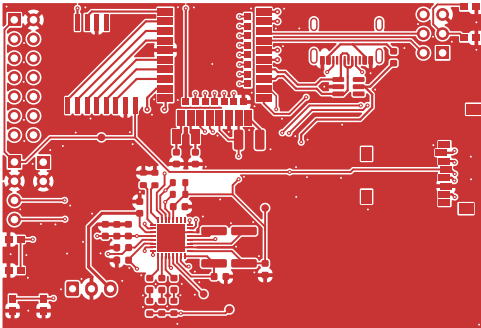


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		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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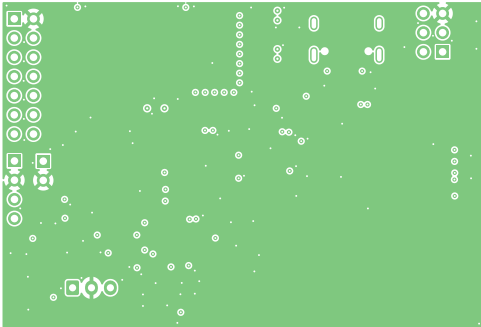
L1 (Sig, PWR) (Scale 1:1)



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		Karpát Space Program		CHECKED	e9cdcd6
	Sheet Title:	Board Name:		Project Name:	
		CanSat Devboard		Cansat Devboard	
	L1 (Sig, PWR) (Scale 1:1)	File Name:	Designer:	Date:	Revision:
		CanSat_Devboard.kicad_pcb	Bertalan Nagy	2024-04-13	+ (Unreleased)
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				A4	7 of 10

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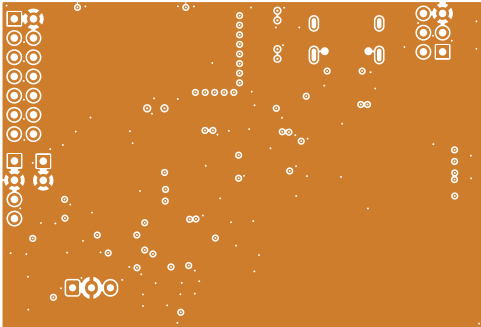
L2 (GND) (Scale 1:1)



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	Sheet Title: L2 (GND) (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
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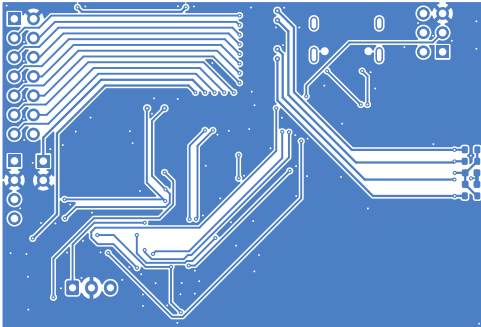
L3 (GND) (Scale 1:1)



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		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
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L4 (Sig, PWR) (Scale 1:1)



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		Board Name: CanSat Devboard		Project Name: Cansat Devboard	
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: CanSat_Devboard.kicad_pcb	Designer: Bertalan Nagy	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 10 of 10